Mobile Computing

Lecture 4

Wireless Communication Issues

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Introduction to Wireless Communication

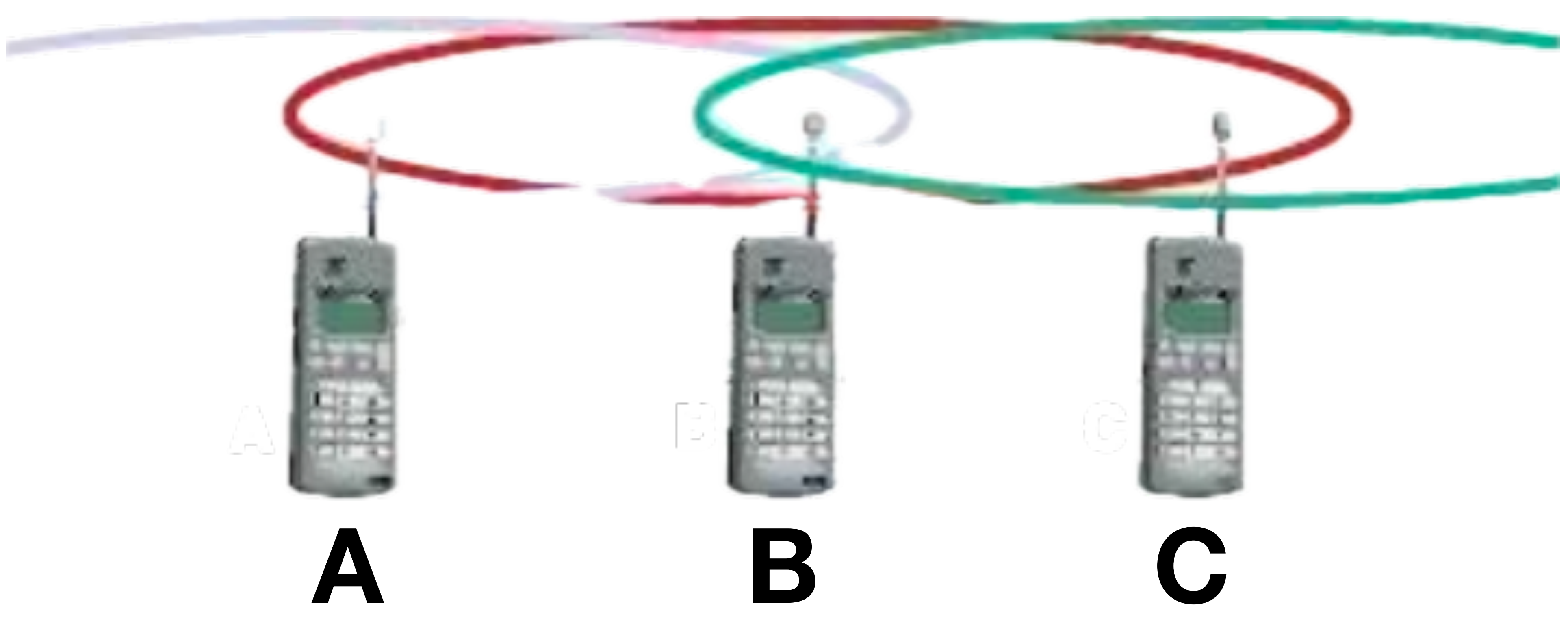
- Wireless communication relies on shared channels.
- Collisions occur due to interference.
- Medium Access Control (MAC) protocols help manage channel access.
- Two key issues: **Hidden Terminal** & **Exposed Terminal** Problems.

Hidden Terminal Problem

Definition

Occurs when two nodes cannot sense each other but interfere with a common receiver.

- **Example**:
    - Node A → Node B
    - Node C → Node B
    - A & C cannot hear each other but both send data to B, leading to collisions.



Hidden Terminal Problem

Causes and Effects

Causes:

- Physical obstructions.
- Weak transmission power.

Effects:

- Increased collisions and packet loss.
- Decreased network efficiency.

Hidden Terminal Problem

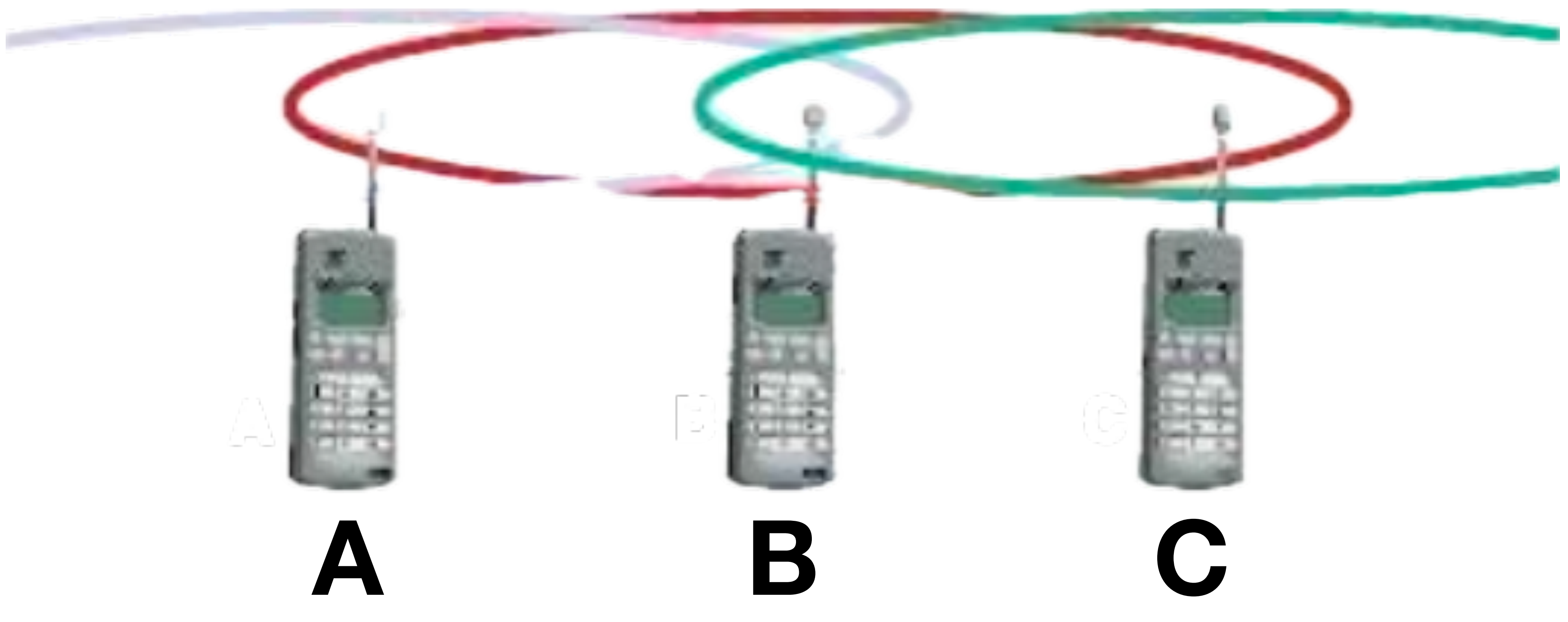
Solutions

- **Solutions:**
  - Use RTS/CTS (Request to Send / Clear to Send) mechanism.
  - Power control techniques.
  - Improved MAC protocols.

Hidden Terminal Problem

Possible Scenario

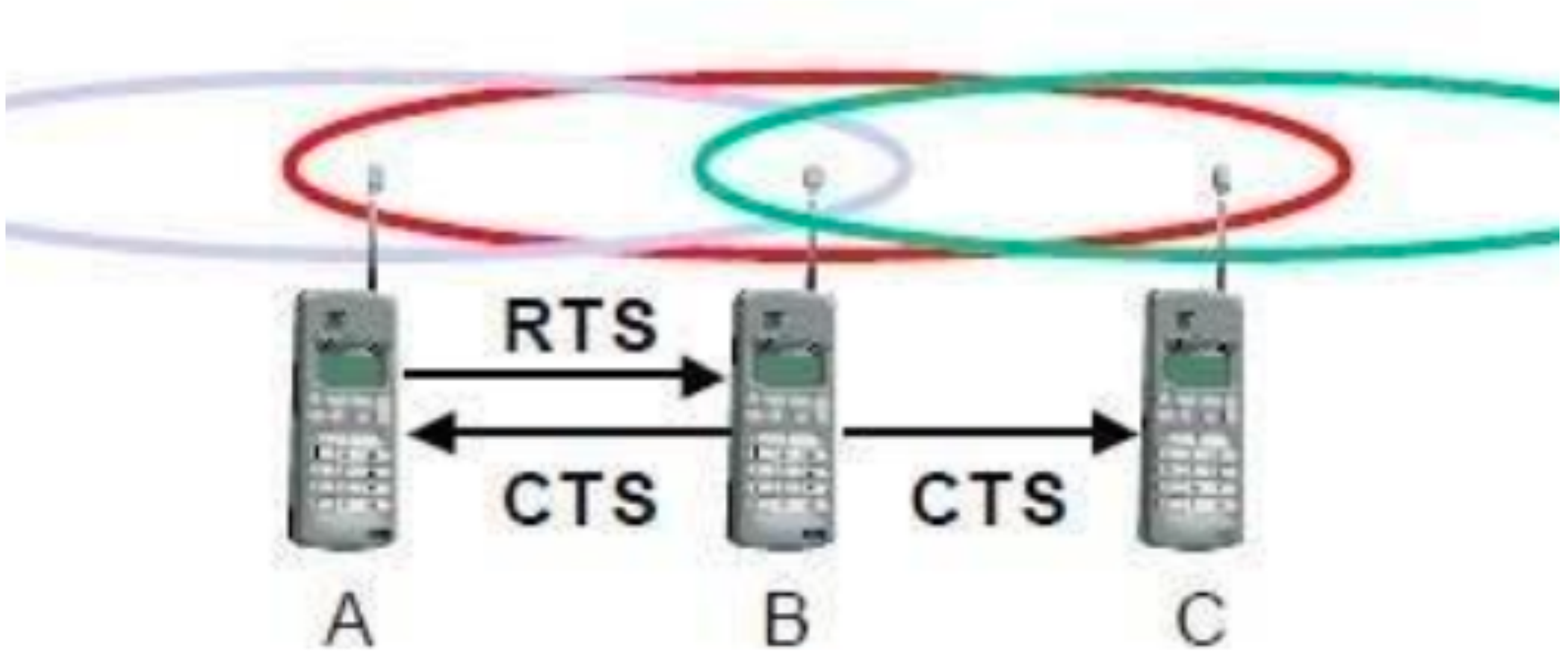
- A starts sending to B, C does not receive this transmission.
- C also wants to send a message to B and senses the medium.
- The medium appears to be free (i.e., the carrier sense fails).
- Thus, C starts sending simultaneously causing a collision at B.
- A can not detect this collision at B and keeps going with its transmission.
- A is hidden for C and vice versa.



Hidden Terminal Problem

RTS/CTS Mechanism

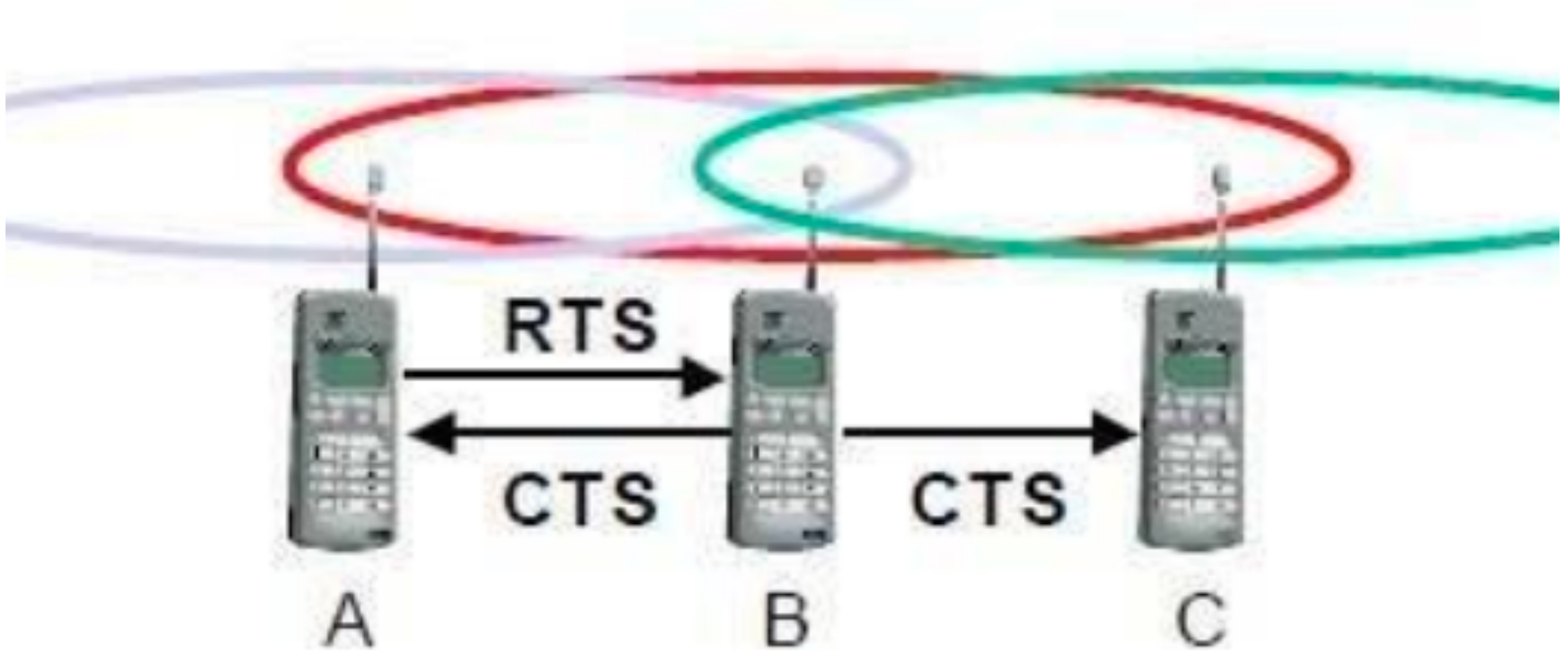
- Multiple access with collision avoidance (MACA) presents a simple scheme that solves the hidden terminal problem.
- With MACA, A doesn't initiate its transmission immediately. Instead, it sends a **request to send** (RTS) first.
- B receives the RTS, which includes the sender's and receiver's names and the length of the upcoming transmission.
- This RTS is not heard by C, but triggers an acknowledgement from B, called **clear to send** (CTS).
- The CTS also contains the sender's (A) and receiver's (B) names and the transmission length.



Hidden Terminal Problem

RTS/CTS Mechanism

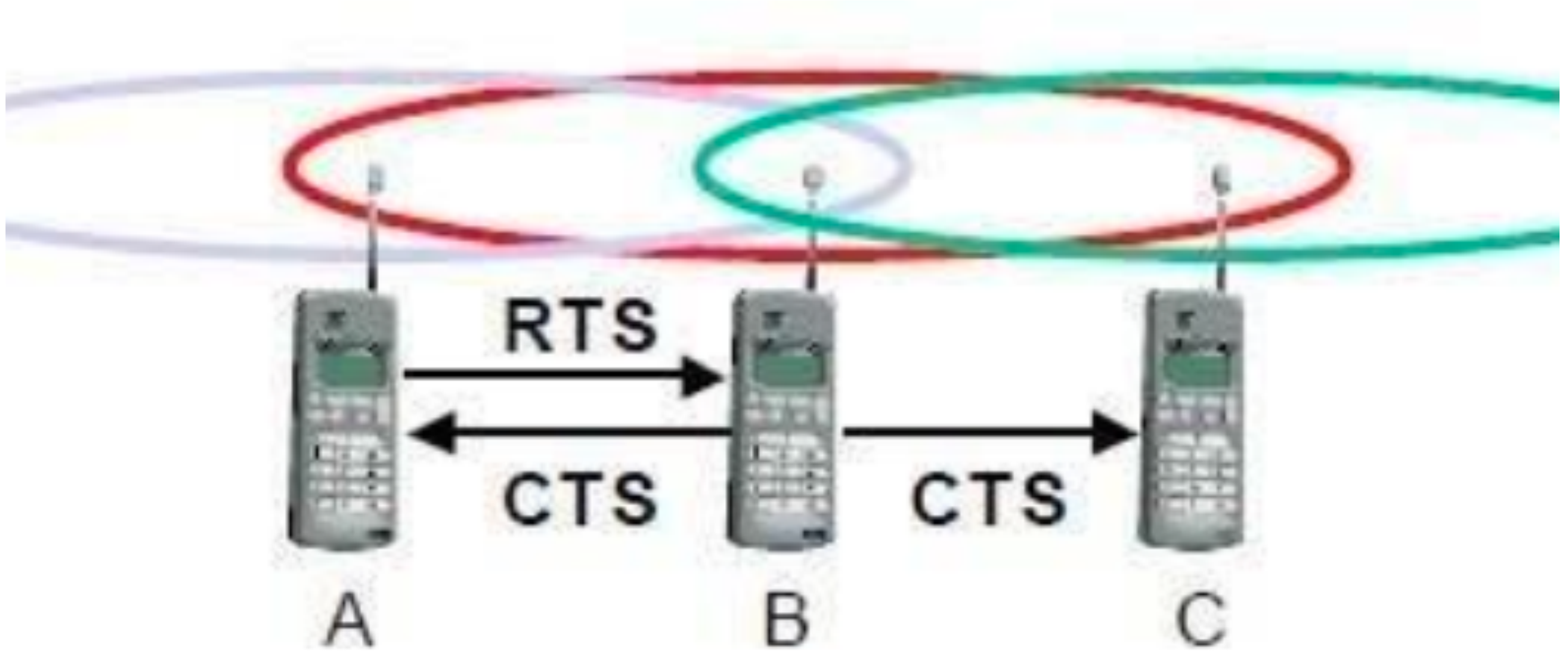
- Any station hearing the RTS is close to the transmitting station and remains silent long enough for the CTS.
- Any station hearing the CTS is close to the receiving station and remains silent during the data transmission.



Hidden Terminal Problem

RTS/CTS Mechanism

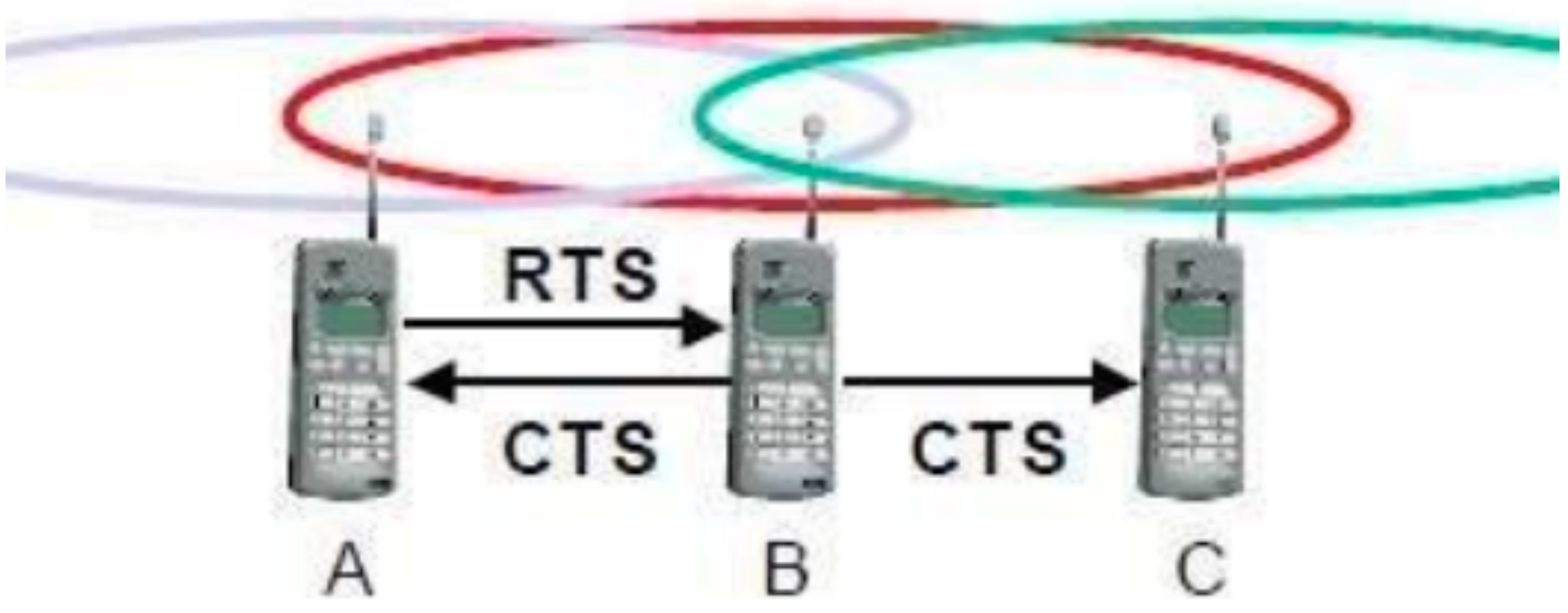
- This CTS is now detected by C and the medium is reserved for future use by A for the specified duration of the transmission.
- Upon receiving a CTS, C is prohibited from sending any data to B for the duration specified in the CTS.
- This eliminates the possibility of a collision at B during data transmission, effectively resolving the hidden terminal problem.



Hidden Terminal Problem

RTS/CTS Mechanism

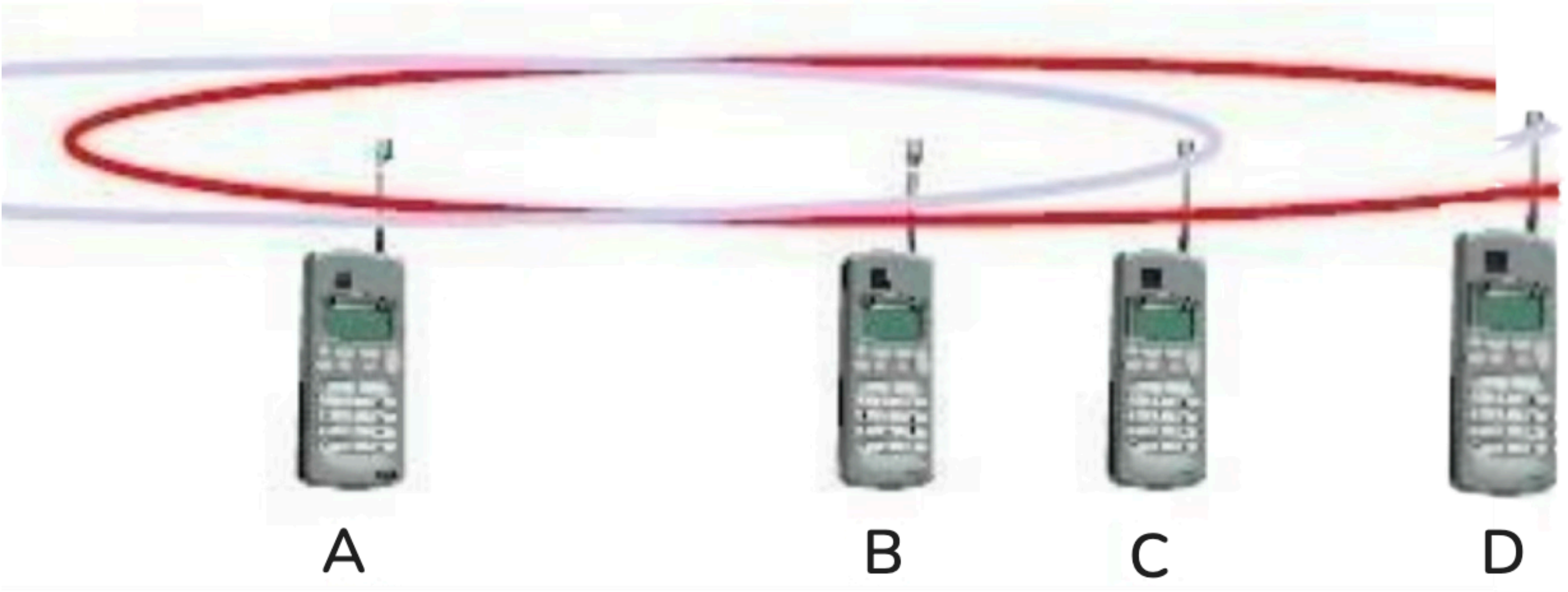
- Still collisions might occur when A and C transmits an RTS at the same time.
- In such situation, B resolves this contention and acknowledges only one station in the CTS.
- No transmission is allowed without an appropriate CTS.



Exposed Terminal Problem

Definition

Occurs when a node unnecessarily refrains from transmitting, though it would not cause interference.



- **Example**:
  - Node B → Node A (transmitting)
  - Node C → Node D (wants to transmit)
  - Transmitters (B&C) are in radio range of each other.
  - Recievers (A&D) are out of radio range of each other.
  - Node C senses B’s transmission and stops sending to D, even though it would not interfere. Why? Because communication from C to D is out of range of Node A.

Exposed Terminal Problem

Causes and Effects

Causes:

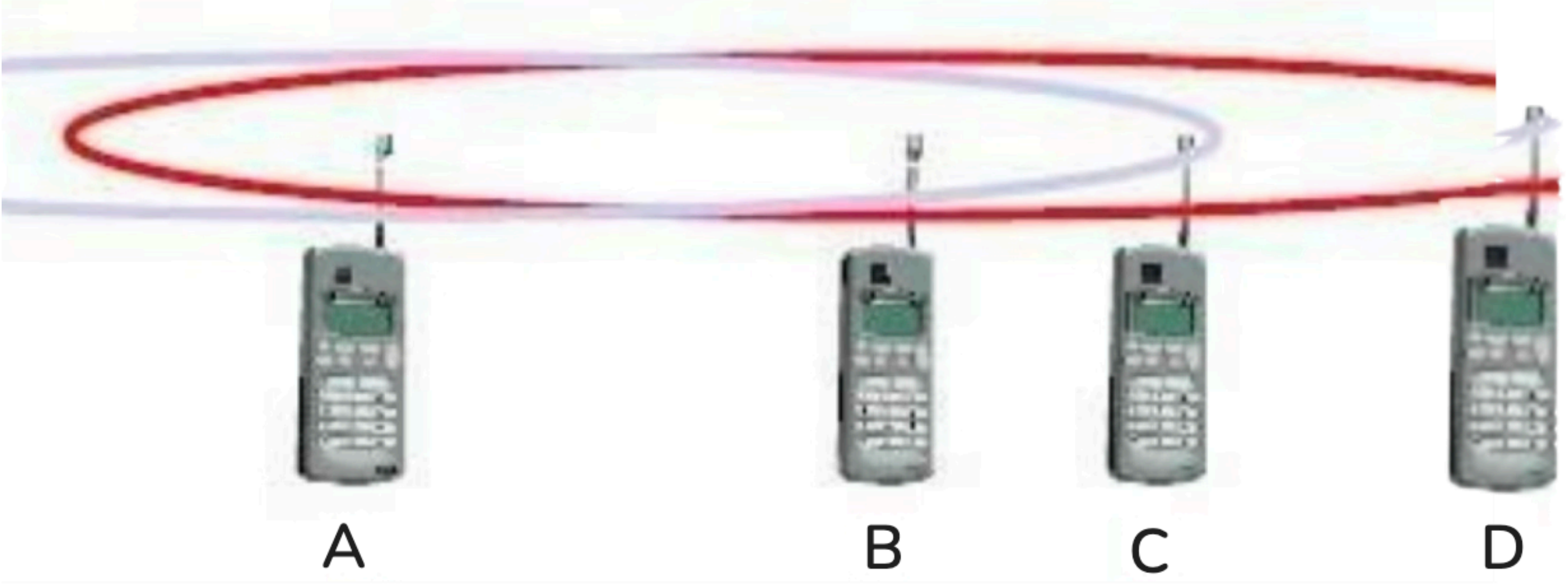
Carrier sensing misinterpretation.

MAC protocol limitations.

Effects:

Underutilization of bandwidth.

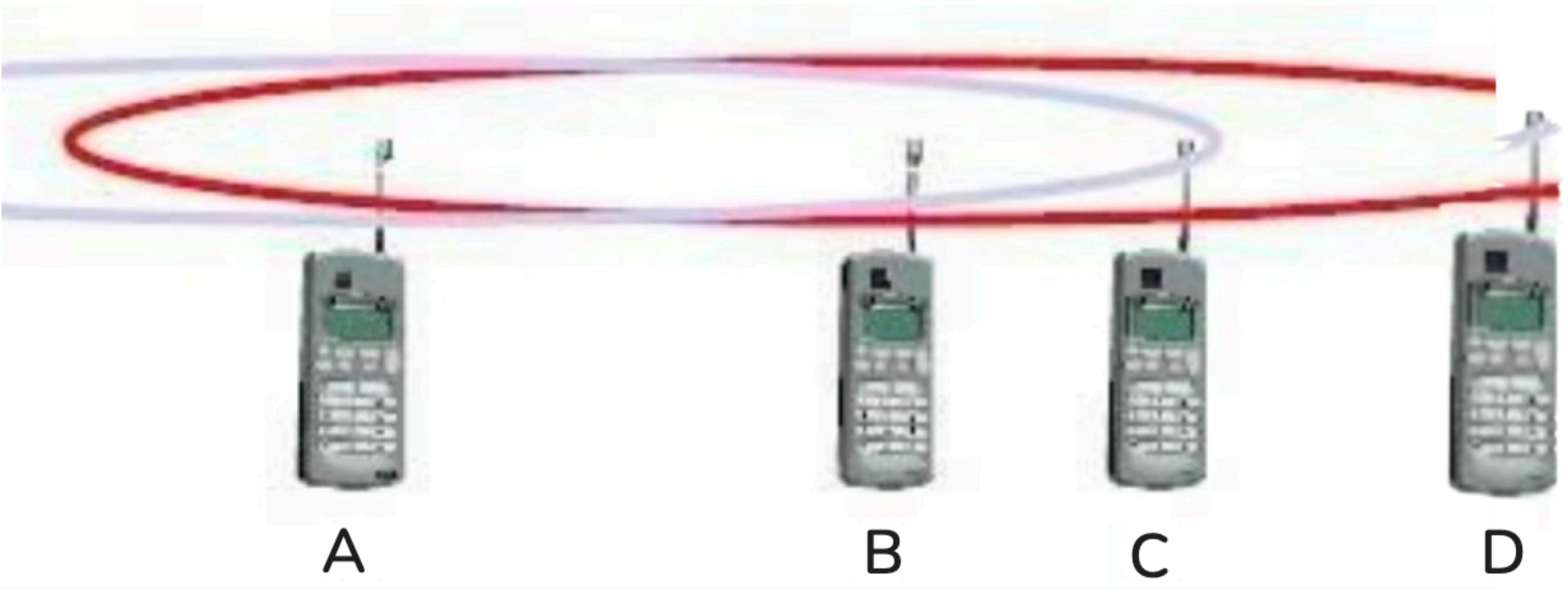
Reduced network throughput.



Exposed Terminal Problem

Solutions

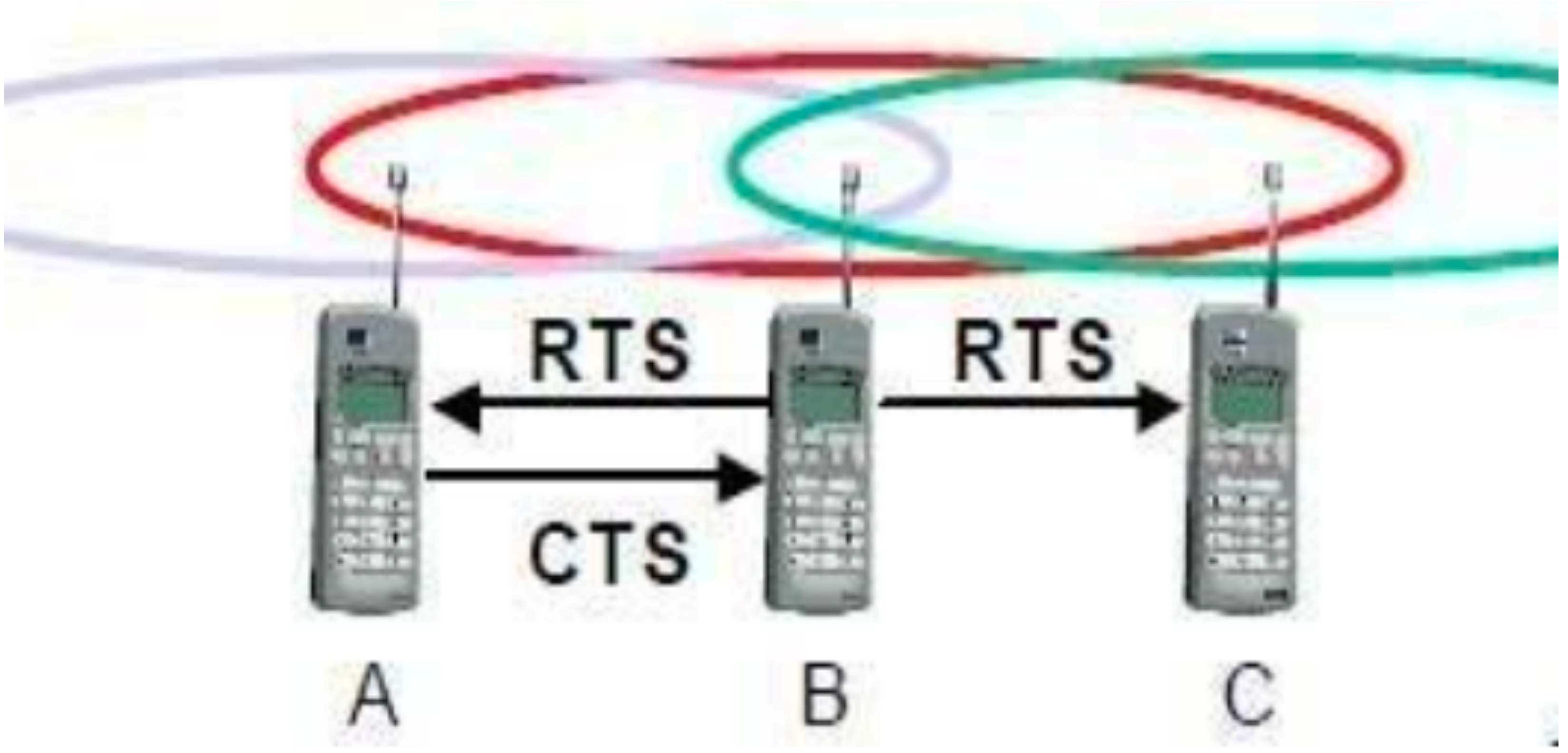
- RTS/CTS mechanism.
- Adaptive sensing techniques.
- Better MAC designs.



Exposed Terminal Problem

RTS/CTS Mechanism

- **MACA Protocol**: B transmits RTS, A responds with CTS, C detects A is outside detection range since it does not receive the CTS from A. Thus, C can start its transmission assuming it will not cause a collision at A.



- **Exposed Terminal Problem**: Solved without fixed access patterns or a base station.

- **RTS/CTS Mechanism**: B identifies receiver (A) and sender (B) in RTS, A acknowledges with CTS identifying sender and receiver.

Hidden and Exposed Terminal Problems

A Comparison

Feature	Hidden Terminal Problem	Exposed Terminal Problem
Issue Type	Collision Issue	Unnecessary Transmission Blocking
Cause	Nodes can't sense each other	Nodes wrongly assume interference
Effect	High packet loss	Reduced network efficiency
Solution	RTS/CTS, power control	RTS/CTS, adaptive sensing

Conclusion

- Hidden and exposed terminal problems affect wireless communication efficiency.
- RTS/CTS and improved MAC protocols help mitigate these issues.
- Understanding these problems leads to better network performance.

Questions?